Installation Manual

Student Course Insight Chatbot

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1. System Requirements

Before installing, ensure the following are available on the host machine:

Requirement	Details
Docker Desktop	Version 24.0 or later (includes Docker Compose v2)
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+
RAM	Minimum 4 GB (8 GB recommended)
Disk Space	Minimum 3 GB free
Internet Connection	Required for Docker image pull and OpenAI API calls
OpenAI API Key	Required for embedding generation and LLM responses

2. Files Required

The following files must be present in the project root directory:

- docker-compose.yml — orchestrates all three services
- .env — environment variables (created from .env.example)
- CommentsData06032026a.xlsx — source feedback dataset
- backend/ — FastAPI backend source code
- frontend/ — React frontend source code

3. Step-by-Step Installation

Step 1 — Install Docker Desktop

Download and install Docker Desktop from:

```
https://www.docker.com/products/docker-desktop
```

After installation, verify Docker is running:

```
docker --version
docker compose version
```

Step 2 — Obtain the Project Files

Clone the repository or extract the submitted ZIP file:

```
git clone <repository-url>
cd finaldemo
```

Step 3 — Configure Environment Variables

Copy the example environment file:

```
cp .env.example .env        (macOS / Linux)
copy .env.example .env      (Windows)
```

Open .env in a text editor and set:

```
DATABASE_URL=postgresql://insight:insight_pass@localhost:5432/insight_db
SECRET_KEY=replace-with-a-long-random-secret-string
OPENAI_API_KEY=sk-...your-openai-api-key-here...
```

NOTE: The OPENAI_API_KEY must be a valid key from https://platform.openai.com. Without it, data ingestion and query generation will not function.

Step 4 — Build and Start All Services

Run the following command from the project root directory:

```
docker compose up --build
```

This command will:

1. Pull the PostgreSQL + pgvector Docker image
2. Build the FastAPI backend image
3. Build the React frontend image
4. Start all three containers
5. Automatically ingest feedback data on first run (requires OpenAI API key)

The first build may take 3-5 minutes depending on network speed.

Step 5 — Verify the System is Running

Once startup is complete, open a browser and navigate to:

Service	URL
Frontend Application	http://localhost:5173
Backend API	http://localhost:8000
API Documentation (Swagger)	http://localhost:8000/docs
Health Check	http://localhost:8000/health

The health check should return: {"status": "ok"}

Step 6 — Log In

The following demo accounts are seeded automatically:

Email	Password	Role
user@example.com	password123	Standard User
admin@example.com	password123	Administrator

4. Stopping and Restarting

Action	Command
Stop all services	docker compose down
Stop and remove all data (full reset)	docker compose down -v
Restart after stopping	docker compose up
Rebuild after code changes	docker compose up --build

5. Running Tests

All tests use mocked dependencies and run fully offline — no database or API key required.

Backend Tests (pytest)

```
cd backend
pip install pytest pytest-cov
pytest tests/ -v --cov=app --cov-report=term-missing
```

Frontend Tests (Vitest)

```
cd frontend
npm install
npm test
```

6. Troubleshooting

Problem	Cause	Solution
Frontend shows blank page	Backend not yet ready	Wait 30 seconds and refresh
Query generation fails	Invalid or missing OPENAI_API_KEY	Add valid key to .env, then: docker compose restart backend
Privacy gate triggers on all queries	Auto-ingest failed (no data in DB)	Run: docker exec -it insight_backend python scripts/ingest.py --file /app/CommentsData06032026a.xlsx
Port 5173 or 8000 already in use	Another service on same port	Edit docker-compose.yml, change host port (left side of colon)
Cannot delete user with history	Old DB schema without CASCADE	Run: docker compose down -v && docker compose up --build
Backend container exits immediately	PostgreSQL not ready in time	Run: docker compose restart backend

7. Local Development Setup (Without Full Docker)

If Docker is not available for the application, only the database can be run in Docker:

Start database only

```
docker compose up db -d
```

Start backend

```
cd backend
pip install -r requirements.txt
python scripts/ingest.py --file ../CommentsData06032026a.xlsx
python run.py
```

Start frontend (separate terminal)

```
cd frontend
npm install
```

```
npm run dev
```

Service	URL
Backend API	http://localhost:8000
Frontend	http://localhost:5173

8. Security Notes

- Never commit the .env file to version control (excluded by .gitignore)
- The SECRET_KEY should be a long, random string in any real deployment
- Demo credentials should be changed before any production use
- The OpenAI API key must only be stored in .env — never hardcoded in source code